DSAA3073: Theories in Data Science

Week 5 Validation Sheet · Stability, Tikhonov, and Ridge

Coverage: Week 5 **Duration:** 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. At a fixed input x_0, which condition is needed to remove a cross term between the fitted value $\hat{f}_{S(x_0)}$ and fresh test noise ε in the squared-loss bias-variance decomposition?

(a) $\mathbb{E}[\varepsilon] = 0$ is always sufficient.
(b) $\mathbb{E}[\varepsilon \mid S, x_0] = 0$, as supplied by fresh zero-mean noise.
(c) The training predictions have zero variance.
(d) The fitted value equals $f^{\star(x_0)}$ for every sample.

Solution. (b). The fitted value is a function of the random training sample S. Conditioning on S gives

$$\mathbb{E}\left[\left(\hat{f}_{S(x_0)} - \mu\right)\varepsilon\right] = \mathbb{E}_S\left[\left(\hat{f}_{S(x_0)} - \mu\right)\mathbb{E}[\varepsilon \mid S, x_0]\right].$$

Hence the conditional zero-mean statement in (b) makes this term vanish. The unconditional equality $\mathbb{E}[\varepsilon] = 0$ in (a) does not rule out correlation with the fitted value. Options (c) and (d) are much stronger properties and are not assumptions of the decomposition.

Marking. *3 marks for (b); 2 for identifying conditional freshness, rather than unconditional zero mean, as the reason the cross term vanishes.*

A2 [5 marks]. Compare $A_0 = \mathrm{diag}(1, 0)$ and $A_\varepsilon = \mathrm{diag}(1, \varepsilon)$ with $\varepsilon > 0$. Which statement is correct on $\mathbb{R}^2$?

(a) Both have a discontinuous inverse.
(b) A_0 has a unique inverse, while A_ε may have none.
(c) A_0 can fail existence or uniqueness; A_ε is invertible but has inverse gain $\frac{1}{\varepsilon}$ in its second direction.
(d) A small nonzero singular value is mathematically identical to a zero one.

Solution. (c). The zero second singular value of A_0 collapses a whole input direction. An observation outside its range has no exact preimage, while a range observation can have infinitely many preimages. For $\varepsilon > 0$, A_ε is invertible in finite dimensions, but an observation perturbation in the second direction is multiplied by $\frac{1}{\varepsilon}$. Thus a zero direction and a small positive direction have qualitatively different diagnoses.

Marking. *3 marks for (c); 2 for distinguishing loss of information at zero from large but finite amplification when $\varepsilon > 0$.*

A3 [5 marks]. Why does the finite-dimensional objective $J_{\lambda(f)} = \|Af - y\|_2^2 + \lambda\|f\|_2^2$ have one global minimizer for every matrix A when $\lambda > 0$?

(a) Every convex problem has a unique minimizer.
(b) $A^T A + \lambda I$ is positive definite.
(c) The residual can always be made zero.
(d) The penalty removes the noise from y.

Solution. (b). For every nonzero vector z,

$$z^{T(A^T A + \lambda I)} z = \|Az\|_2^2 + \lambda\|z\|_2^2 > 0.$$

Hence the Hessian $2(A^TA + \lambda I)$ is positive definite, so the quadratic is strictly convex and its stationary point is the unique global minimizer. Convexity alone does not imply uniqueness, an exact fit need not exist, and regularization does not remove observation noise.

Marking. *3 marks for (b); 2 for the positive-definite quadratic-form reason and its uniqueness consequence.*

A4 [5 marks]. In a positive eigendirection of X^TX with eigenvalue μ, what fraction of the OLS coefficient is retained by ridge at $\lambda > 0$?

(a) $\frac{1}{\mu+\lambda}$
(b) $\frac{\lambda}{\mu+\lambda}$
(c) $\frac{\mu}{\mu+\lambda}$
(d) Either 0 or 1, because ridge performs subset selection.

Solution. (c). Ridge changes the directional coefficient from $\frac{z}{\mu}$ to $\frac{z}{\mu+\lambda}$, so their ratio is $\frac{\mu}{\mu+\lambda}$. The quantity in (a) is the inverse factor acting on z, not the fraction of the OLS coefficient. Ridge is continuous spectral shrinkage; it does not generally make the binary keep-or-discard choice described in (d).

Marking. *3 marks for (c); 2 for distinguishing the shrinkage factor from the inverse factor and from subset selection.*

Part B: Diagnose a repeated-fit system

B1 [20 marks]. Three systems have the same training MSE. At one test input, $f^\star(x_0) = 4$. The table reports four independent refit predictions, the eigenvalues of the normalized Gram matrix, the fraction of refit movement aligned with the weakest eigendirection, and the fresh-noise standard deviation.

system	refits	spectrum	weak share	σ
P	$3.7, 4.1, 3.9, 4.3$	$(1, 0.7, 0.4)$	20%	0.2
Q	$2.0, 5.5, 2.5, 6.0$	$(1, 0.5, 0.01)$	90%	0.2
R	$2.8, 3.2, 3.0, 3.0$	$(1, 0.8, 0.6)$	15%	0.2

Diagnose the dominant problem in each system. For each one, defend or reject ridge as the first intervention using at least two columns of evidence. State the most that the Week 5 evidence can justify without choosing a numerical value of λ.

Solution. The columns must be read together; the common training MSE does not decide among the systems.

System P. The refits are centered at 4 because their mean is $\frac{3.7+4.1+3.9+4.3}{4} = 4$, and their spread is small. The spectrum has no extremely weak direction, and only 20 percent of the movement aligns with the weakest direction. There is therefore no strong evidence that poor conditioning is the dominant failure. Ridge is not the first targeted response on these data.

System Q. Its refit mean is also 4, but the predictions vary sharply. The smallest Gram eigenvalue is 0.01, and 90 percent of the movement lies in that weak direction. Fresh-noise scale is the same modest 0.2 as in the other systems. This combination identifies variance amplified through poor conditioning. Ridge is a targeted candidate because it attenuates the weak direction, while its added bias must be monitored.

System R. The refits are tightly clustered around mean 3, one unit away from $f^\star(x_0) = 4$. The spectrum is not weak and little motion lies in its weakest direction. The dominant evidence is systematic bias, not unstable inversion. Increasing squared-norm shrinkage is not a justified first remedy and can increase the displacement; the representation or model is a more relevant object to reconsider.

Week 5 therefore justifies **considering** ridge for Q and explains which failure it addresses. It does not identify a numerical λ and does not establish that predictive risk will fall. Such a choice needs honest out-of-sample evidence or another justified selection criterion introduced later, with a final test set kept untouched.

Marking. *4 marks for diagnosing P from both its small centered cloud and nonweak spectrum; 6 for diagnosing Q from the large aligned refit spread and 0.01 eigenvalue and defending ridge as targeted; 4 for diagnosing R as systematic bias and rejecting stronger shrinkage as the first remedy; 3 for recognizing that the common training MSE and common fresh-noise scale do not settle the diagnosis; 3 for limiting the conclusion to considering ridge without selecting λ or promising lower predictive risk.*

Part C: Basic calculations

C1 [7 marks]. At fixed x_0, four equally likely training samples produce predictions $2, 4, 4, 6$. If $f^{\star(x_0)} = 5$ and the fresh-noise variance is 3, compute the squared bias, training-sample variance, and expected squared prediction error. Use population averaging over the four declared outcomes.

Solution. The mean fitted prediction is

$$\mu = \frac{2+4+4+6}{4} = 4.$$

Hence the squared bias is $\left(\mu - f^{\star(x_0)}\right)^2 = (4-5)^2 = 1$. The variance over the four equally likely training-sample outcomes is

$$\frac{(2-4)^2 + (4-4)^2 + (4-4)^2 + (6-4)^2}{4} = \frac{8}{4} = 2.$$

Under the stated fixed-input squared-loss and fresh-noise assumptions, the decomposition gives expected squared prediction error

$$1 + 2 + 3 = 6.$$

Dividing by four is correct here because the four values are the complete declared distribution, not a sample used to estimate an unknown variance.

Marking. *2 marks for $\mu = 4$ and squared bias 1; 3 for variance 2 with the declared divisor; 2 for total expected error 6 and the three-term interpretation.*

C2 [8 marks]. In one positive eigendirection of $X^T X$, let $\mu = 4$, $\lambda = 1$, the true coefficient be $\theta = 2$, and the noise variance be $\sigma^2 = 3$. Compute the ridge shrinkage factor, the mean ridge coefficient, its bias, and its variance.

Solution. The shrinkage factor is

$$s(\lambda) = \frac{\mu}{\mu + \lambda} = \frac{4}{5}.$$

Therefore the mean ridge coefficient is $s(\lambda)\theta = \left(\frac{4}{5}\right)2 = \frac{8}{5}$ and its bias relative to $\theta = 2$ is

$$\frac{8}{5} - 2 = -\frac{2}{5}.$$

Under the fixed-design homoskedastic model, the directional coefficient variance is

$$\sigma^2 \frac{\mu}{(\mu+\lambda)^2} = \frac{3 \cdot 4}{5^2} = \frac{12}{25}.$$

These are coefficient-space moments; prediction-space contributions would require the additional directional design weight.

Marking. *2 marks for $s = \frac{4}{5}$; 2 for mean $\frac{8}{5}$; 2 for bias $-\frac{2}{5}$; 2 for variance $\frac{12}{25}$.*

Part D: Construct and verify boundary cases

D1 [17 marks]. Consider $A_0 = \mathrm{diag}(2, 0)$ and $A_s = \mathrm{diag}(2, 0.02)$ on $\mathbb{R}^2$.

For A_0, construct one observation with no exact solution and a different observation with infinitely many exact solutions, verifying both claims. For A_s, construct a perturbation of norm 0.002 whose recovered change has norm 0.1. Use the constructions to explain why a zero singular value and a small positive singular value are not the same failure.

Solution. The construction should place all decisive behavior in the second coordinate.

For nonexistence under A_0, take $y = (0, 1)$. Every forward image has the form

$$A_0(f_1, f_2) = (2f_1, 0),$$

whose second coordinate is zero. Hence no f can satisfy $A_0 f = y$.

For nonuniqueness, take $y' = (2, 0)$. The equation requires $f_1 = 1$ but places no restriction on f_2, so every vector $(1, t)$ with $t \in \mathbb{R}$ is an exact solution. Thus the zero singular direction contains information that the forward map completely discards.

The matrix A_s is invertible. Choose the observation perturbation $\Delta y = (0, 0.002)$. Its Euclidean norm is 0.002, while

$$\Delta f = A_s^{-1}\Delta y = \left(0, \frac{0.002}{0.02}\right) = (0, 0.1),$$

which has norm 0.1. The amplification factor is $50 = \frac{1}{0.02}$.

Consequently, A_0 may fail existence or uniqueness because information is absent. In contrast, A_s has a unique solution and a continuous inverse in finite dimensions, but is quantitatively badly conditioned: small data changes can be enlarged by a large finite factor.

Marking. *4 marks for a valid no-solution observation and complete range check; 4 for a valid infinitely-many-solutions observation and parameterized solution family; 4 for constructing Δy and obtaining Δf with the required norms; 3 for the 50 amplification interpretation; 2 for the correct zero-versus-small qualitative distinction. Equivalent coordinate-scaled constructions receive full credit.*

D2 [18 marks]. Let

$$A = \mathrm{diag}\left(2, \frac{1}{2}\right), \quad y = (4, 1), \quad \lambda = 1,$$

and $J(f) = \|Af - y\|_2^2 + \|f\|_2^2$. Derive the unique minimizer over all $f \in \mathbb{R}^2$. Then verify, rather than merely assert, that it is the global minimizer. Finally compare its two coordinates with those of the exact inverse $A^{-1}y$.

Solution. The Tikhonov normal equations are

$$(A^T A + I)f = A^T y.$$

Here

$$A^T A + I = \mathrm{diag}\left(5, \frac{5}{4}\right), \quad A^T y = \left(8, \frac{1}{2}\right).$$

Solving coordinatewise gives

$$f_\lambda = \left(\frac{8}{5}, \frac{\frac{1}{2}}{\frac{5}{4}}\right) = \left(\frac{8}{5}, \frac{2}{5}\right).$$

To verify globality, take any nonzero z. The Hessian satisfies

$$z^T \nabla^2 J(f) z = 2\|Az\|_2^2 + 2\|z\|_2^2 > 0.$$

Therefore J is strictly convex on all of $\mathbb{R}^2$. Its stationary point is consequently the unique global minimizer, not merely a local candidate.

Exact inversion gives

$$A^{-1}y = \left(\frac{4}{2}, \frac{1}{\frac{1}{2}}\right) = (2, 2).$$

Ridge therefore retains the fractions $\frac{\frac{8}{5}}{2} = \frac{4}{5}$ and $\frac{\frac{2}{5}}{2} = \frac{1}{5}$ in the two directions. The weaker second direction is shrunk more strongly. This is continuous shrinkage; neither coordinate comparison is a subset-selection decision.

Marking. *4 marks for forming the correct normal equations and matrices; 4 for solving $f_\lambda = (\frac{8}{5}, \frac{2}{5})$; 5 for a valid positive-definite or completing-the-square proof of unique global optimality; 3 for computing $A^{-1}y = (2, 2)$ and the retained fractions; 2 for interpreting stronger shrinkage in the weaker direction.*

Part E: Transfer to a coupled inverse

E1 [25 marks]. Let

$$v_+ = \frac{1, 1}{\sqrt{2}}, \quad v_- = \frac{1, -1}{\sqrt{2}}, \quad u_1 = (1, 0), \quad u_2 = (0, 1).$$

These are orthonormal bases of the input and observation spaces. For $\tau \in \{0, 0.1\}$, a linear map is defined by

$$A_\tau v_+ = 2u_1, \quad A_\tau v_- = \tau u_2.$$

The observed vector is $y = 2u_1 + 0.2u_2$. For $\lambda > 0$, define

$$J_{\tau,\lambda}(f) = \|A_\tau f - y\|_2^2 + \lambda \|f\|_2^2.$$

Determine the unique minimizer $f_{\tau,\lambda}$, expressed in standard coordinates, for each value of τ. Use your result to compare the two maps' treatment of the u_2 observation component and to determine both limits as $\lambda \to 0$. Explain precisely which limit is an exact solution and which is the Moore–Penrose minimum-norm least-squares solution.

Solution. Choose a representation that exposes the map. Write an arbitrary input as $f = av_+ + bv_-$. Orthonormality gives

$$\|f\|_2^2 = a^2 + b^2, \quad A_\tau f = 2au_1 + \tau bu_2.$$

Since u_1, u_2 are also orthonormal, the objective separates as

$$J_{\tau,\lambda}(a, b) = (2a - 2)^2 + (\tau b - 0.2)^2 + \lambda(a^2 + b^2).$$

This change of representation is the key step: in standard coordinates the map is coupled, whereas in the supplied bases the two directional decisions become visible.

Differentiating the separated quadratic gives

$$(4 + \lambda)a = 4, \quad (\tau^2 + \lambda)b = 0.2\tau.$$

Hence, for either declared τ,

$$a_\lambda = \frac{4}{4 + \lambda}, \quad b_{\tau,\lambda} = 0.2\frac{\tau}{\tau^2 + \lambda}.$$

The Hessian in (a, b) coordinates is $2\,\mathrm{diag}(4 + \lambda, \tau^2 + \lambda)$, which is positive definite because $\lambda > 0$. The stationary point is therefore the unique global minimizer.

Standard coordinates follow from $av_+ + bv_- = \left(\frac{a+b}{\sqrt{2}}, \frac{a-b}{\sqrt{2}}\right)$. For $\tau = 0$,

$$b_{0,\lambda} = 0, \quad f_{0,\lambda} = \frac{4}{4 + \lambda}v_+ = \frac{1}{\sqrt{2}}\left(\frac{4}{4 + \lambda}, \frac{4}{4 + \lambda}\right).$$

The u_2 data component cannot affect the solution because it lies outside range(A_0). It remains as the unavoidable residual $0.2u_2$, while the zero-prior penalty sets the unobservable v_- coefficient to zero.

For $\tau = 0.1$,

$$b_{0.1,\lambda} = \frac{0.02}{0.01 + \lambda},$$

so

$$f_{0.1,\lambda} = \frac{1}{\sqrt{2}}\left(\frac{4}{4+\lambda} + \frac{0.02}{0.01+\lambda}, \frac{4}{4+\lambda} - \frac{0.02}{0.01+\lambda}\right).$$

The u_2 direction is now active but weak. Its regularized data-to-solution gain is $\frac{0.1}{0.01+\lambda}$ rather than the unregularized gain 10, so positive λ attenuates its influence without declaring it absent.

Taking $\lambda \to 0$ through positive values gives

$$f_{0,\lambda} \to v_+, \quad f_{0.1,\lambda} \to v_+ + 2v_-.$$

For $\tau = 0.1$, direct substitution yields $A_{0.1}(v_+ + 2v_-) = 2u_1 + 0.2u_2 = y$, so the limit is the unique exact inverse solution. For $\tau = 0$, no exact solution exists because y has a nonzero u_2 component outside the range. All least-squares minimizers have the form $v_+ + tv_-$: they fit the u_1 component and leave the same residual $0.2u_2$. The limit selects $t = 0$, the least-norm member. It is therefore $A_0^\dagger y$, the Moore–Penrose minimum-norm least-squares solution, not an ordinary inverse of A_0.

Marking. *4 marks for independently choosing the v_+, v_- representation and correctly separating the objective; 4 for deriving the two directional equations; 3 for proving unique global optimality; 4 for the correct standard-coordinate formulas for both τ values; 3 for interpreting the absent versus attenuated u_2 influence; 3 for both zero-penalty limits; 4 for proving that the 0.1 limit is exact while the zero case selects the minimum-norm member of the full least-squares family. Preserve method and follow-through credit after an isolated arithmetic slip.*

Part F: Limit a claim before selecting a parameter

F1 [15 marks]. An analyst increases λ and observes two facts: the minimized training residual becomes larger, and the spread of predictions across repeated fits becomes smaller. They conclude, "test error must therefore have decreased, so these observations are enough to choose this value of λ."

Audit the conclusion using only Week 5 concepts. State what the two facts do establish, identify the missing quantities, and explain why Week 5 diagnosis alone cannot honestly select a numerical λ.

Solution. The two observations establish a real but limited tradeoff. A larger training residual means the penalized fit gives up some in-sample agreement. A smaller refit spread is evidence that estimator variance has fallen for the inspected predictions. This is consistent with ridge damping sensitive directions.

Neither fact determines test error. Shrinkage also displaces the estimator's mean, so squared bias may increase. The predictive effect depends on where the true signal lies relative to the spectral directions and on how test inputs weight those directions. Prediction of fresh responses additionally includes the fresh-noise variance, which regularization does not remove. Thus the relevant risk balances squared bias, variance, and irreducible noise; it need not be monotone in λ.

The training residual is not an out-of-sample risk estimate, and a refit cloud at selected inputs does not supply the missing full predictive-risk comparison. Week 5 can diagnose that variance or poor conditioning makes ridge worth considering and can predict which directions will be damped. It cannot certify the best numerical penalty from those facts alone. Honest selection requires later out-of-sample evidence or another justified criterion, while the final test set remains unused until the procedure is fixed. Therefore "must have decreased" and "enough to choose" are both unsupported.

Marking. *3 marks for correctly interpreting the larger residual; 3 for correctly interpreting reduced refit spread as variance evidence; 5 for identifying added bias, spectral/test-input weighting, and fresh-noise limitations; 2 for rejecting monotonicity of test error; 2 for stating that honest later selection evidence is required and the final test set cannot be used for tuning.*